

Advancing Legal Reasoning in Algerian Law

Integrating **RAG**, **Knowledge Graphs** & **Argument Mining** for **Citation-Faithful** Legal Reasoning

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Defended **13 / 06 / 2026**

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The Problem

WHY LEGAL AI IS HARD

Legal answers cannot be judged by fluency. They must be **grounded in identifiable sources**, carry **valid article-level citations**, stay inside the **right jurisdiction**, respect legal structure and time — and **abstain** when the law is silent.

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Where fluent LLMs fail in law

- Hallucinated citations
- Attribution gap
- Foreign-law contamination
- No abstention
- Retrieval blindness
- Multi-hop fragility

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Algerian law is a demanding case

- Predominantly Arabic
- Historically layered 1963–2025
- Structurally complex
- Low-resource in legal AI

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Research gap

No Algerian-law system jointly handles article-level citation faithfulness, amendment-aware structure, jurisdictional discipline and principled abstention. Scaling the model alone does not fix it.

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Objective

Build **AKN-RLM** — an architecture that grounds every answer in **verifiable Algerian legal sources**, and declines when the corpus does not justify an answer.

171

CORPUS FILES

189MB

MULTIMODAL

244

BENCHMARK QS

Two novel resources: a structured AKN corpus + the **AlgerianLegalBench** benchmark.

2

The AKN-RLM Pipeline

DOCUMENTS → STRUCTURE → VERIFIED ANSWER

A heterogeneous architecture where each layer repairs a specific failure mode of the last — not one model, but a controlled evidence pipeline.

— STRUCTURE THE LAW —

**Raw Algerian Legal Texts**

PDF · TXT sources, 1963–2025

1

**Akoma Ntoso XML Corpus**

Article segmentation · FRBR registry

2

**Legal Knowledge Graph**

RDF · amendment chains · cross-refs

3



— RETRIEVE & REASON —

**Hybrid Retrieval**

BM25 + Dense, RRF fusion

4

**Typed-Handler Dispatch**

Classifier routes 8 query types

5

**Bounded Recursive Reasoning**

Gap-probe RLM + Toulmin argument mining

6



— VERIFY & ANSWER —

**Triple Faithfulness Gate**

Citation existence · jurisdiction · claim support

7

**Verifiable Answer or Abstention**

Every claim resolves to a registered article

8

3

Evaluation & Results

FROZEN 244-QUESTION BENCHMARK

23

CATEGORIES

8QUERY
TYPES**40**INFECTION
TRAPS**.829**K
AGREEMENT**0.305****Citation F1**
locked deployable
classifier-typed**0.000**

HCR · JIR

0.703

Abstention F1

📊

Citation F1 vs. baseline tiers

Direct LLM		0.056
raw, no retrieval		
Deterministic		0.105
hybrid + rerank		
Minimal RAG		0.175
dense top-5		
Strongest LLM		0.186
Gemini 2.5 Flash		
AKN-RLM		0.305
locked deployable		

1.64×vs direct
LLM**2.9×**vs determin-
istic retrieval**1.74×**vs minimal
RAG

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Citation faithfulness findings

The **only** system to hold both hallucinated-citation and jurisdictional-infection rates at **0.000**. Gemini 2.5 Flash still imports foreign law on **15 of 40** traps (JIR 0.375); Qwen 2 falls into **27** (JIR 0.675).

Reliability tracks **legal structure, not parametric scale** — a smaller controller leads, while the largest baselines are the most hallucination-prone.

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Limitations

- Retrieval ceiling: Recall@10 = 0.258 (gold in top-10 on ~¼ of queries)
- French under-served — Arabic-primary system
- Abstention precision 0.654
- Metrics are proxies, not expert review

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Future work

- Corpus-tuned dense embedder
- French ↔ Arabic retrieval
- Graph-retrieval redesign
- Faithfulness-gate ablations
- Expert legal validation

**THE TAKEAWAY**

Trustworthy legal AI must be built as an **evidence-controlled system**, not an unconstrained generator — it requires **structured legal sources**, controlled retrieval, recursive reasoning, and **strict citation verification**.